

Product rule

Assume $b > 0$, $b \neq 1$, $M > 0$, and $N > 0$.

$$\log_b (MN) = \log_b M + \log_b N$$

Which argument operation becomes addition?

Expanding a product

Assume $x > 0$.

$$\log_2(8x)$$

What equivalent sum uses a numerical constant?

Quotient rule

Assume $b > 0$, $b \neq 1$, $M > 0$, and $N > 0$.

$$\log_b \left(\frac{M}{N} \right) = \log_b M - \log_b N$$

Which argument operation becomes subtraction?

Expanding a quotient

Assume $x > 0$.

$$\ln\left(\frac{x}{e^2}\right)$$

What equivalent difference uses a numerical constant?

Power rule

Assume $b > 0$, $b \neq 1$, $M > 0$, and p is real.

$$\log_b (M^p) = p \log_b M$$

What happens to the exponent p ?

Expanding a power

Assume $x > 0$.

$$\log_5 (x^4)$$

What equivalent expression uses the power rule?

Expanding a radical

Assume $x > 0$.

In $\sqrt{x}$

What equivalent expression has no radical?

Expanding a negative power

Assume $x > 0$.

$$\log_3 (x^{-2})$$

What equivalent expression has no exponent inside the logarithm?

Several factors

Assume $x > 0$ and $y > 0$.

$$\ln(7x^2y)$$

What equivalent sum has no products inside logarithms?

A product and a quotient

Assume $x > 0$ and $y > 0$.

$$\log_2 \left(\frac{8x^3}{y^2} \right)$$

What fully expanded expression results?

Factoring before expanding

Assume $x > 2$.

$$\ln(x^2 - 4)$$

What equivalent sum follows from factoring the argument?

No rule for a sum

$$\ln(x + y) \quad \ln x + \ln y$$

Are these expressions equal for every positive x and y ?

Condensing a sum

Assume $x > 0$.

$$\log_3 x + \log_3 9$$

What single logarithm is equivalent?

Condensing a difference

Assume $x > 0$.

$$\ln x - \ln 5$$

What single logarithm is equivalent?

Condensing with coefficients

Assume $x > 0$ and $y > 0$.

$$2\ln x - 3\ln y$$

What single logarithm is equivalent?

A fractional coefficient

Assume $x > 0$.

$$\frac{1}{2}\log x + \log 4$$

What single common logarithm is equivalent?

Domain during expansion

$$\ln(x^2) \quad 2\ln x$$

Do these expressions have the same real domain?

An absolute-value correction

$$\ln(x^2) = 2\ln|x|$$

For which real x is this identity defined?

Domain during condensation

$$\ln(x - 1) + \ln(x + 1)$$

What original domain must accompany the condensed expression?

Change of base

Assume $b > 0$, $b \neq 1$, and $M > 0$.

$$\log_b M = \frac{\ln M}{\ln b}$$

Why is the denominator nonzero?

A calculator-ready logarithm

$$\log_7 20$$

What is the value rounded to four decimal places?

A reciprocal relationship

$$\log_2 7 \quad \log_7 2$$

What is the product of these two values?